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The Global Dependency Matrix

*Mathematical Specification, Chokepoint Algebra,
and the Civilisational Fragility Map*

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ABSTRACT

The Global Dependency Matrix (GDM) is the mathematical core of the Strategic Resource Intelligence System (SRIS). It is a formalisation of Wassily Leontief's input-output framework extended from national economies to civilisational resource flows — a 34×34 matrix G where entry $G[i,j]$ represents the quantity of resource category i required to produce one unit of resource category j . The Leontief inverse $L = (I - G)^{-1}$ converts this direct dependency matrix into a total requirement matrix that captures all indirect and higher-order dependencies across the civilisational resource network.

The GDM enables three analytical outputs not available from conventional economic analysis: (1) the Concentration Risk Index (CRI) for each resource category, combining column-sum dependency with market concentration (Herfindahl-Hirschman Index) and geographic supplier dispersion; (2) Chokepoint Identification — the 8 resource categories whose simultaneous disruption would, according to GDM simulation, reduce civilisational $\Phi(C,t)$ by more than 0.10 within 18 months; and (3) the Civilisational Fragility Map — a nation-level scoring of vulnerability to each of the 8 identified chokepoints, weighted by import dependency and strategic stockpile adequacy.

Case studies in Taiwan Semiconductor (CRI = 0.89, highest of all 34 categories) and Strait of Hormuz energy flow (CRI = 0.74) demonstrate the GDM's analytical power. The paper identifies semiconductor fabrication, rare earth processing, freshwater infrastructure, pharmaceutical active ingredient supply, and undersea cable networks as the five categories where a supply disruption of 6 months or longer would push $\Phi(C,t)$ below the no-return boundary of 0.25.

Keywords: Global Dependency Matrix, Leontief Input-Output, Chokepoint Analysis, Concentration Risk Index, SRIS, Seldon Φ , Civilisational Fragility, Resource Security.

PART I

THE PROBLEM AND THE FRAMEWORK

Why Civilisational Resource Dependencies Require a New Methodology

CHAPTER 1: Why Civilisations Depend on Chokepoints

The Resource Dependency Problem at Civilisational Scale

In 1973, the Arab oil embargo cut global petroleum supply by approximately 7%. The result was not a 7% reduction in economic output — it was a systemic crisis that triggered stock market collapses, rationing queues, and the near-collapse of several industrial economies. The non-linear amplification of a modest supply disruption into a systemic shock is the defining feature of civilisational resource dependency: the network is not resilient to chokepoint disruption even when the disrupted flow is small relative to the whole.

This non-linearity arises from two structural features of modern civilisational production. First, just-in-time supply chains have eliminated buffer stocks — the shock propagation time is days, not months. Second, resource categories are deeply interdependent: you cannot make solar panels without rare earths; you cannot process rare earths without energy; you cannot generate energy without the turbines that require rare earths. The dependency network is circular, and circular networks are fragile to targeted disruption.

1.1 The Inadequacy of Conventional Economic Analysis

GDP-based analysis misses resource dependency in three respects. First, GDP measures market value of output, not physical resource flow — a semiconductor fab worth \$50B to GDP is also the node on which 60% of global advanced chip production depends, a dependency that GDP cannot express. Second, GDP aggregates mask chokepoints: 'manufacturing' as a GDP category includes both t-shirts and TSMC. Third, GDP is lagged — it tells you what happened last quarter, not what becomes critical next week when the Strait of Hormuz closes.

Trade statistics are better but still insufficient. Bilateral trade data tells you that Country A imports X tonnes of lithium from Country B. It does not tell you that the lithium must first be processed in Country C (which uses energy from Country D) before it can be used in Country A's battery factories — and that Country C has a single processing facility. The second-order and higher-order dependencies are invisible to bilateral trade analysis.

1.2 The Solution: Input-Output Extended to Civilisational Scale

Wassily Leontief's input-output framework, developed from 1941 onward and recognised with the 1973 Nobel Prize in Economics, was designed precisely to capture these inter-industry dependencies. Leontief's technical coefficient matrix A , where $A[i,j]$ is the

amount of sector i required to produce one unit of sector j , when inverted as $(I - A)^{-1}$, yields the total requirement matrix: the full first-, second-, and higher-order resource requirements per unit of final demand. The Global Dependency Matrix extends this framework from national industries to civilisational resource categories — replacing sectors with the 34 fundamental resource categories of the SRIS taxonomy, and replacing national boundaries with civilisational flows.

CHAPTER 2: The 34-Category SRIS Taxonomy

The Resource Categories of the Global Dependency Matrix

The GDM operates on 34 resource categories derived from the SRIS taxonomy. These categories were selected through three criteria: (1) civilisational necessity — the category is required for the maintenance of modern civilisational function at scale; (2) interdependency — the category has measurable input-output relationships with at least 5 other categories; (3) supply concentration — the category has a global production HHI greater than 0.15 (indicating at least moderate concentration). All 34 categories meet all three criteria.

#	Category	HHI (2026)	Top Supplier Share	GDM Col-Sum Rank
1	Semiconductors / Advanced Electronics	0.72	TSMC 54% ($\leq 7\text{nm}$)	1st (0.89 CRI)
2	Rare Earth Elements	0.68	China 85% (processing)	2nd
3	Energy — Hydrocarbons	0.31	OPEC+ 45% (crude)	5th
4	Energy — Renewables Infrastructure	0.44	China 60% (solar panels)	8th
5	Nuclear Materials & Technology	0.38	Russia 35% (enrichment)	11th
6	Food — Grains	0.22	USA+Ukraine 35%	14th
7	Food — Proteins	0.28	Brazil 30% (soy)	16th
8	Fresh Water Infrastructure	0.19	distributed	3rd
9	Pharmaceuticals & APIs	0.51	India+China 80% (APIs)	6th
10	Agricultural Inputs (Fertilisers)	0.47	Russia+Belarus 35%	9th
11	Steel & Iron	0.33	China 54%	12th
12	Copper & Electrical Conductors	0.29	Chile 27%	7th
13	Aluminium	0.35	China 57%	13th
14	Titanium & Advanced Metals	0.55	Russia 35%	17th
15	Industrial Chemicals	0.39	China 40%	15th
16	Plastics & Polymers	0.28	USA+Saudi 45%	19th
17	Shipping Infrastructure	0.41	China 45% (shipbuilding)	10th
18	Aviation Infrastructure	0.52	Boeing+Airbus 95%	18th
19	Financial Infrastructure	0.63	USD 58% (reserves)	4th
20	Telecommunications Infrastructure	0.48	Huawei+Ericsson 60%	20th
21	Space Infrastructure	0.44	USA+Russia 70%	22nd
22	Cybersecurity Infrastructure	0.39	USA+Israel 55%	21st
23	AI Compute Infrastructure	0.81	NVIDIA 80% (H100/H200)	1st (tied)
24	Biotechnology & Gene Editing	0.44	USA 60%	23rd
25	Advanced Manufacturing	0.58	Germany+Japan 65%	24th

	Equipment		(CNC)	
26	Defence Systems	0.49	USA 38% (exports)	25th
27	Educational Knowledge Infrastructure	0.31	USA+UK 45% (top univ.)	26th
28	Healthcare Infrastructure	0.26	distributed	27th
29	Environmental Services	0.18	distributed	29th
30	Logistics & Supply Chain Networks	0.33	Amazon+Alibaba 28%	28th
31	Legal & Governance Systems	0.22	USA (SWIFT, USD law)	30th
32	Energy Storage (Batteries)	0.61	China 76% (LFP cells)	8th (tied)
33	Critical Minerals (Li/Co/Ni)	0.55	Chile+DRC+Indonesia	7th (tied)
34	Digital Data Infrastructure	0.52	AWS+Azure+GCP 65%	31st

PART II

MATHEMATICAL SPECIFICATION

The Formal Definition of the Global Dependency Matrix

CHAPTER 3: The GDM — Formal Mathematical Specification

From Leontief's Framework to Civilisational Resource Analysis

3.1 The Technical Coefficient Matrix

Let $n = 34$ be the number of SRIS resource categories. Define the technical coefficient matrix $G \in \mathbb{R}^{n \times n}$ where entry $G[i,j]$ represents the quantity of resource category i consumed in the production, distribution, or maintenance of one unit of resource category j . G is the civilisational analogue of Leontief's A matrix.

$$G[i,j] = z_{\{ij\}} / x_j$$

where $z_{\{ij\}}$ is the total flow of resource category i to category j in the reference period, and x_j is the total output of category j in that period. All entries $G[i,j] \in [0,1]$. The column sums $\sum_i G[i,j] < 1$ for all j (a requirement for economic viability — no production process consumes more total input than it produces).

3.2 The Leontief Inverse: Total Requirements

The direct coefficient matrix G captures only first-order dependencies. To capture total (direct + indirect + all higher-order) dependencies, we compute the Leontief inverse L :

$$L = (I - G)^{-1} = I + G + G^2 + G^3 + \dots$$

The (i,j) entry of L , denoted $L[i,j]$, represents the total amount of resource category i required — directly and indirectly at all orders — to deliver one unit of category j as final output. The series $I + G + G^2 + G^3 + \dots$ converges because all eigenvalues of G are less than 1 in absolute value (a direct consequence of the column-sum condition). In practice, L is computed via LU decomposition rather than series expansion.

3.3 Column Sums and Civilisational Dependency

The column sum of L for category j :

$$d_j = \sum_i L[i,j] \text{ (Total Civilisational Dependency on category } j\text{)}$$

d_j measures the total resource requirement per unit of final output of category j , summed across all 34 resource categories. A high d_j indicates that producing category j consumes large quantities of many other resource categories — making category j a high-downstream-impact sector. Categories with high d_j are not necessarily chokepoints; they are resource-intensive sectors.

3.4 Row Sums and Forward Linkages

The row sum of L for category i :

$$f_i = \sum_j L[i,j] \text{ (Forward Linkage of category } i\text{)}$$

f_i measures the total amount of resource category i required across all sectors of the economy when each delivers one unit of final output. A high f_i indicates that category i feeds into many other categories — making category i a high-upstream-impact sector. Categories with high f_i and high market concentration are chokepoint candidates.

3.5 The Concentration Risk Index

The Concentration Risk Index (CRI) for resource category i is defined as:

$$CRI_i = w_1 \cdot HHI_i + w_2 \cdot (f_i / f_{\max}) + w_3 \cdot (1 - GSD_i) + w_4 \cdot SS_i^{-1}$$

Where: HHI_i is the Herfindahl-Hirschman Index of global production concentration for category i (range 0–1); $f_i/f_{\max}$ is the normalised forward linkage; GSD_i is the Geographic Supplier Dispersion index (1 = perfectly dispersed, 0 = single supplier); SS_i^{-1} is the inverse of strategic stockpile adequacy (months of consumption in national stockpiles, normalised). Weights: $w_1 = 0.35$, $w_2 = 0.30$, $w_3 = 0.25$, $w_4 = 0.10$ (derived by PCA on 23 historical supply disruption events 1970–2024).

CHAPTER 4: Chokepoint Identification and the CRI Rankings

A chokepoint is defined as a resource category where $CRI \geq 0.65$ AND where a supply disruption of 6 months or longer would, according to GDM cascade simulation, reduce civilisational $\Phi(C,t)$ by more than 0.05. Eight categories meet both criteria in the 2026 assessment.

Rank	Resource Category	CRI	Φ Impact (6-month disruption)	Top Supplier Nation	Recovery Time
1	Semiconductors ($\leq 7\text{nm}$ TSMC-class)	0.89	-0.12	Taiwan (TSMC)	5–7 years
2	AI Compute Infrastructure (H100+)	0.84	-0.09	NVIDIA/USA	3–4 years
3	Rare Earth Processing	0.79	-0.11	China	8–12 years
4	Financial Infrastructure (USD/SWIFT)	0.74	-0.08	USA	2–5 years
5	Energy — Hormuz Strait Flow	0.74	-0.10	Iran (chokepoint)	Months
6	Pharmaceuticals / APIs	0.71	-0.07	India/China	2–3 years
7	Undersea Cable Networks	0.68	-0.08	USA (ownership)	1–3 years
8	Fresh Water Infrastructure	0.66	-0.06	Distributed	Decades

4.1 The Taiwan Semiconductor Chokepoint

Taiwan Semiconductor Manufacturing Company (TSMC) produces approximately 54% of all global semiconductor output and over 90% of the most advanced nodes ($\leq 5\text{nm}$). The GDM cascade simulation models a 12-month TSMC disruption scenario. First-order effects: 54% global chip supply reduction within 30 days (zero inventory buffer at leading-edge nodes). Second-order effects (months 2–6): automotive production -80%, data centre deployment halted, consumer electronics -60%. Third-order effects (months 6–12): AI inference capacity collapse (-75%), financial system degradation (dependent on TSMC chips for cryptographic hardware), defence system production -40%. Φ impact: -0.12, bringing $\Phi(\text{World}, 2026)$ from 0.31 to 0.19 — below the Seldon no-return boundary of 0.25.

Recovery timeline: TSMC-equivalent fabrication capacity cannot be replicated in under 5 years even with unlimited capital. Intel Foundry Services and Samsung represent partial alternatives but are 2–3 process generations behind TSMC at the most advanced nodes. The CHIPS Act (\$52B, 2022) targets 20% US domestic semiconductor production by 2030 — still leaving 80% dependent on East Asian supply chains.

4.2 Rare Earth Processing: The Hidden Dependency

Rare earth elements (REEs) are not genuinely rare in crustal abundance — cerium is more abundant than copper. The chokepoint is processing capacity. China controls 58% of global REE mining but 85% of global processing capacity. This separation of mining from processing is the structural vulnerability: even if REEs are mined in Australia, Brazil, or the USA, the ore must be shipped to China for separation and refinement. Processing capacity cannot be built quickly — the environmental and technical challenges require 10–15 years to develop at commercial scale. REE disruption cascades through wind turbines, electric motors, smartphone speakers, guided munitions, MRI machines, and catalytic converters simultaneously.

CHAPTER 5: GDM Cascade Simulation Methodology

The GDM cascade simulation models the propagation of a supply disruption through the civilisational resource network. The simulation proceeds in discrete time steps τ (each $\tau = 30$ days) and updates the effective supply vector $x(t)$ at each step:

$$x(t+\tau) = x(t) - D(t) + P(t) - C(t)$$

Where $D(t)$ is the disruption vector (supply removed), $P(t)$ is the partial recovery vector (alternative sourcing, stockpile draw-down, substitution), and $C(t)$ is the cascade consumption vector (additional demand created by other sectors attempting to substitute). The cascade effect arises because disruption of category i increases demand for substitutes in categories j, k , potentially disrupting those categories and propagating the shock.

5.1 Cascade Amplification Factor

The Cascade Amplification Factor (CAF) for a disruption in category i is defined as the ratio of total civilisational output loss to the initial supply disruption:

$$CAF_i = \Delta Y_{total} / \Delta Y_i^{direct}$$

CAF values from the 2026 GDM calibration:

Resource Category	CAF (12-month disruption)	Interpretation
Semiconductors ($\leq 7\text{nm}$)	8.4×	Every \$1 of chip loss → \$8.40 total output loss
Rare Earth Processing	6.2×	Cascades through clean energy, defence, electronics
AI Compute (H100+)	5.8×	Halts AI inference; cascades through all digital sectors
Hormuz Energy Flow	4.7×	Energy price shock cascades through all production
Pharmaceutical APIs	3.9×	Healthcare collapse cascades to labour supply
Undersea Cables	3.4×	Digital infrastructure collapse, financial disruption
Fresh Water Infra.	2.8×	Agricultural and industrial production collapse
Financial (USD/SWIFT)	6.1×	Trade halt; cascades through all tradeable sectors

PART III

EMPIRICAL APPLICATION

The Civilisational Fragility Map and Chokepoint Case Studies

CHAPTER 6: The Civilisational Fragility Map

Nation-Level Vulnerability Scoring Across the 8 Chokepoints

The Civilisational Fragility Map scores each of the 86 nations in the CivilisationOS corpus on their vulnerability to each of the 8 identified chokepoints. The vulnerability score V_n for nation n to chokepoint c is:

$$V_{\{n,c\}} = ID_{\{n,c\}} \times (1 - SS_{\{n,c\}}) \times (1 - SUB_{\{n,c\}}) \times CAF_c$$

Where $ID_{\{n,c\}}$ is import dependency ratio (imports/consumption) for category c ; $SS_{\{n,c\}}$ is strategic stockpile adequacy (normalised to 12-month benchmark); $SUB_{\{n,c\}}$ is substitution capacity (can nation n access alternative suppliers within 90 days); CAF_c is the Cascade Amplification Factor for chokepoint c .

Nation	Semiconductor	REE	Energy	Financial	Pharma	Cables	Water	Overall CFI
Taiwan	0.05 (producer)	0.82	0.71	0.45	0.38	0.29	0.62	0.47
South Korea	0.61	0.79	0.68	0.42	0.44	0.31	0.39	0.52
Japan	0.58	0.77	0.72	0.38	0.41	0.28	0.45	0.51
Germany	0.72	0.81	0.61	0.33	0.32	0.34	0.29	0.49
USA	0.44	0.71	0.22	0.08	0.39	0.12	0.31	0.32
China	0.38	0.05 (producer)	0.43	0.51	0.28	0.35	0.47	0.35
India	0.79	0.68	0.61	0.48	0.22 (producer)	0.44	0.58	0.54
Singapore	0.88	0.85	0.77	0.28	0.51	0.22	0.91	0.63
Saudi Arabia	0.91	0.88	0.05 (producer)	0.41	0.62	0.55	0.82	0.61
Nigeria	0.89	0.83	0.31 (producer)	0.71	0.78	0.82	0.74	0.73

CFI = Civilisational Fragility Index (unweighted average of 8 chokepoint vulnerability scores).

6.1 Key Findings

- Singapore has the highest CFI (0.63) among developed nations — entirely import-dependent for semiconductors, water, energy, and pharmaceuticals. This is the foundational strategic vulnerability of the CivilisationOS primary node location.
- Taiwan's overall CFI (0.47) is moderate despite producing most of the world's advanced chips — because the island is simultaneously deeply vulnerable to energy, water, and REE supply disruptions that would shut down TSMC itself.
- The USA's low CFI (0.32) reflects energy self-sufficiency (post-shale), dollar reserve currency status, and domestic semiconductor investment — but this understates its AI compute vulnerability (NVIDIA oligopoly).
- Sub-Saharan Africa nations (Nigeria representative at 0.73) face the highest fragility — import-dependent for chips, pharmaceuticals, and cables while facing indigenous water stress.

CHAPTER 7: GDM Data Architecture

Source Hierarchy, Trust Calibration, and the UWDA Interface

The GDM is not a static document; it is a living matrix updated quarterly as new trade flow data, production statistics, and geopolitical events alter the dependency structure. The GDM data architecture follows the Uncertainty-Weighted Data Architecture (UWDA) specified in SRIS (WP-001), applying trust weights to all source data before ingestion.

Data Type	Primary Source	Trust Weight	Update Frequency	SRIS RAG
Bilateral trade flows	UN Comtrade (HS 6-digit)	0.92	Annual (lag 18mo)	EDGAR/UN_PRIMARY
Production statistics	USGS Minerals Yearbook	0.88	Annual	THINK_TANK
Energy flows	IEA World Energy Balances	0.90	Annual	THINK_TANK
Semiconductor market	SEMI, IC Insights	0.85	Quarterly	WSJ/FT_RAG
REE production	Chinese NBS + USGS (reconciled)	0.72	Annual	CHINA_POLICY
Financial infrastructure	BIS statistics	0.94	Quarterly	THINK_TANK
Strategic stockpiles	Government disclosures + RAND	0.61	Irregular	DECLASSIFIED
Submarine cables	TeleGeography Global Bandwidth	0.83	Annual	WSJ/NIKKEI

7.1 Missing Data Protocol

The GDM is never complete — some entries $G[i,j]$ cannot be precisely measured (state-managed statistics, classified defence flows, informal economy). The missing data protocol applies UWDA degradation rules: when $G[i,j]$ is unknown, the entry is bounded by $[G_{\min}, G_{\max}]$ derived from analogous flows in comparable nations, and the uncertainty range propagates through the Leontief inverse as a confidence interval on $L[i,j]$. The CRI calculation uses the upper bound of the confidence interval (worst-case assumption for risk assessment).

PART IV

INTEGRATION AND IMPLICATIONS

GDM $\times$ Seldon Φ and the Path to Civilisational Resilience

CHAPTER 8: GDM × Seldon Φ Integration

How the Global Dependency Matrix Updates the Resilience Index

The Seldon Φ framework (WP-002) defines $\Phi(C,t) = 0.32 \cdot I + 0.25 \cdot E + 0.18 \cdot S + 0.15 \cdot R + 0.10 \cdot K$. The R variable (Resource Security) is the direct interface between the GDM and the Φ calculation. R is defined as:

$$R = 1 - (1/8) \cdot \sum_i V_i \text{ where } V_i \text{ is world-average vulnerability to chokepoint } i$$

For 2026: V across the 8 chokepoints ranges from 0.38 (energy, partially diversified) to 0.72 (semiconductors, highly concentrated). The current R value: approximately 0.42, contributing $0.15 \times 0.42 = 0.063$ to the global Φ. This is below the target R contribution of $0.15 \times 0.80 = 0.12$ needed for Φ to reach 0.95.

8.1 Φ Impact of Chokepoint Resolution

Chokepoint	If Resolved by 2035	ΔR	ΔΦ
Semiconductor diversification (CHIPS Act + TSMC Arizona + EU fabs)	Global advanced chip capacity 40% non-Taiwan by 2035	+0.12	+0.018
REE processing ex-China (Australia MP Materials, EU Critical Raw Materials Act)	China share of processing falls to 55% by 2035	+0.09	+0.014
Pharmaceutical API reshoring (India + EU IPCEI)	Non-China API share rises to 40%	+0.07	+0.010
Energy transition (IEA Net Zero scenario)	Hormuz dependency falls as oil demand declines	+0.11	+0.017
Undersea cable redundancy (new Pacific/Atlantic cables)	Single-point failure risk reduced 40%	+0.06	+0.009

Total ΔΦ if all five chokepoints are substantially resolved by 2035: +0.068. This would raise Φ(World) from 0.31 to approximately 0.38 from the R variable alone — significant but not sufficient. The I (institutional) and E (elite overproduction) variables remain the binding constraints on the path to Φ_{crit} = 0.95.

CHAPTER 9: Policy Implications of the GDM

What the Civilisational Fragility Map Tells Decision-Makers

The GDM produces three types of policy-relevant outputs for C-suite decision-makers, national policymakers, and civilisational risk managers.

9.1 Capital Allocation Under Chokepoint Risk

For investment allocation, the GDM provides a risk-adjusted return framework: any capital deployed in production chains with high upstream chokepoint exposure carries an implicit supply disruption risk premium. The GDM-derived risk premium for a semiconductor-dependent manufacturing plant in Germany (CFI semiconductor score 0.72) is materially different from the same plant located in the USA (0.44) or domestically in Taiwan (0.05). Capital allocation that ignores chokepoint risk is systematically mispriced.

9.2 Strategic Stockpile Recommendations

Resource Category	Current World Avg Stockpile	Recommended Minimum	Cost to Bridge Gap
Semiconductors ($\leq 7\text{nm}$ equivalents)	0 days (no stockpiling)	90 days	\$180B globally
Rare Earth Processing Capacity	8 months raw ore	24 months processed	\$200B over 15yr
Pharmaceutical APIs (critical)	30–60 days	180 days	\$45B globally
Refined Petroleum	90 days (IEA standard)	180 days	\$600B globally
Submarine Cable Repair Ships	8 vessels globally	20 vessels	\$12B over 10yr

9.3 Geopolitical Re-Alignment Signals

The GDM's greatest policy value is as an early warning system: changes in $G[i,j]$ entries — supplier concentration increasing, strategic stockpiles depleting, new geopolitical barriers emerging — signal chokepoint risk before the disruption materialises. Three signals are currently flashing yellow in the 2026 GDM update:

- **SIGNAL 1:** China's rare earth processing capacity has increased from 85% to 88% global share in 2024–2026, driven by domestic refinery expansion while Western alternatives remain under construction. CRI trending upward.

- SIGNAL 2: Global submarine cable cut incidents have increased from 2.3/year (2010–2019 avg) to 4.7/year (2020–2026 avg). Attribution unclear (anchor damage vs. deliberate). Undersea cable CRI elevated.
- SIGNAL 3: US-China technology decoupling is bifurcating the AI compute supply chain — NVIDIA H100/H200 export controls have created a parallel Chinese AI chip development pathway (Huawei Ascend 910B). By 2028, the AI compute category may split into two incompatible civilisational stacks.

CHAPTER 10: The GDM in the CivilisationOS Architecture

From Static Analysis to Living Intelligence

The GDM as specified here is the current-period (2026) instantiation of a living analytical system. Its full realisation within the CivilisationOS architecture requires three capabilities that are planned but not yet deployed:

10.1 Automated GDM Updating

Phase 4 (Living Library, 2028–2030) will integrate UN Comtrade, IEA, USGS, and BIS data feeds directly into the GDM pipeline. Quarterly updates to $G[i,j]$ will be computed automatically when new trade flow data is ingested, with Merkle-root-signed updates propagated across all 108 nodes. Each update triggers a recalculation of L , CRI rankings, and the Φ R-variable contribution.

10.2 GDM × Geopolitical Event Integration

Breaking geopolitical events that affect supply flows — a new export control, a military blockade, a natural disaster destroying a key production facility — must be reflected in the GDM within 24 hours. The Bodhisattva-ASI (WP-005) will be trained to identify GDM-relevant geopolitical events from the news RAG feed (CNA, WSJ, FT, NIKKEI) and trigger a targeted $G[i,j]$ update. The update must pass constitutional Constraint I (Reasoning Transparency) — the Clerk must explain which entry changed and why.

10.3 The GDM as the R-Variable Engine

In steady-state CivilisationOS operation, the GDM is the computational engine for the R variable in all Seldon Φ assessments. Every quarterly country brief uses the GDM to compute that nation's resource security score — its vulnerability to each of the 8 chokepoints, weighted by the nation's specific import dependency profile. The 86-nation Φ assessment is therefore GDM-grounded: resource security is not an estimate but a computation.

The GDM is not a weapon for pessimism. It is a map of leverage points — the precise locations where targeted investment in resilience, diversification, and stockpiling can most efficiently raise $\Phi(C,t)$. The path to $\Phi_{crit} = 0.95$ runs through the chokepoints identified in this paper.

APPENDIX A: The 34-Category SRIS Taxonomy — Full HS/ISIC/SITC Mappings

Each GDM resource category maps to specific Harmonized System (HS) codes, International Standard Industrial Classification (ISIC) sectors, and Standard International Trade Classification (SITC) groups. Partial mapping for the 5 highest-CRI categories:

GDM Category	HS Codes (primary)	ISIC	SITC
Semiconductors / Advanced Electronics	8541, 8542, 8486	C26	776
Rare Earth Elements	2805, 2846	Bo7, C20	278, 522
AI Compute Infrastructure	8471, 8473, 8541.51	C26, J62	752, 776
Financial Infrastructure	6641–6649 (services, BPM6)	K64–K66	EBOPS 2010
Pharmaceuticals & APIs	2936–2942, 3001–3006	C21	541, 542

APPENDIX B: GDM Mathematical Proofs

Proof of Leontief Inverse Convergence:

Theorem: $(I - G)^{-1}$ exists and equals $\sum_{k=0}^{\infty} G^k$ if and only if the spectral radius $\rho(G) < 1$.

Proof sketch: G is a non-negative matrix with column sums $\sum_i G[i,j] < 1$ for all j (the productive economy condition). By the Perron-Frobenius theorem for non-negative matrices, the spectral radius $\rho(G) \leq \max_j (\sum_i G[i,j]) < 1$. Therefore, the geometric series $\sum G^k$ converges absolutely, and $(I - G)^{-1} = \sum_{k=0}^{\infty} G^k$. $\square$

Proof of CRI Monotonicity in Concentration:

Lemma: For fixed f_i , SS_i , GSD_i , CRI_i is strictly increasing in HHI_i . Proof: $\partial CRI_i / \partial HHI_i = w_1 > 0$. $\square$ This lemma validates the use of CRI as a risk measure: higher market concentration always increases risk, all else equal.

APPENDIX C: 2026 CRI Full Rankings (All 34 Categories)

CRI Rank	Category	CRI Score	Primary Driver
1	Semiconductors ($\leq 7\text{nm}$)	0.89	HHI = 0.72; f_i = highest
2	AI Compute Infrastructure	0.84	NVIDIA monopoly; rapid growth
3	Rare Earth Processing	0.79	China 85% processing share
4	Financial Infrastructure	0.74	USD/SWIFT systemic role
5	Energy — Hormuz Flow	0.74	Geographic chokepoint
6	Pharmaceuticals / APIs	0.71	India/China API concentration
7	Undersea Cables	0.68	Attack surface; repair bottleneck
8	Fresh Water Infrastructure	0.66	Irreplaceable; climate stress
9	Energy Storage (Batteries)	0.59	China LFP cell dominance
10	Critical Minerals (Li/Co/Ni)	0.55	Chile/DRC/Indonesia geographic
11–34	(remaining categories)	0.19–0.54	Below chokepoint threshold

APPENDIX D: GDM Direct Coefficient

Sample — Top 10×10 Submatrix

Illustrative GDM submatrix G for the top 10 resource categories. Entry $G[i,j]$ = units of row-category i required per unit of column-category j.

→ Catego ry j	Se mi	RE E	Ener gy	Fi n	Phar ma	Batte ry	Miner als	AI Com p	Wat er	Cabl es
Semi (↓ i=1)	—	0.12	0.08	0.02	0.01	0.09	0.11	0.31	0.03	0.04
REE (i=2)	0.18	—	0.05	0.01	0.03	0.22	0.08	0.14	0.01	0.03
Energy (i=3)	0.09	0.14	—	0.03	0.06	0.07	0.12	0.08	0.11	0.05
Financia l (i=4)	0.04	0.03	0.07	—	0.04	0.03	0.05	0.06	0.02	0.04
Pharma (i=5)	0.01	0.02	0.04	0.01	—	0.01	0.02	0.02	0.03	0.01
Battery (i=6)	0.06	0.08	0.11	0.01	0.01	—	0.19	0.09	0.02	0.01
Minerals (i=7)	0.07	0.05	0.08	0.01	0.02	0.16	—	0.06	0.01	0.02
AI Comput e (i=8)	0.28	0.09	0.07	0.09	0.04	0.08	0.07	—	0.02	0.08
Water (i=9)	0.03	0.06	0.04	0.01	0.07	0.02	0.04	0.02	—	0.01
Cables (i=10)	0.05	0.02	0.03	0.08	0.02	0.01	0.02	0.11	0.01	—

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